09-12 第2篇

阅读理解（说明文） Why Do We Have Two Eyes Instead of One?

① If you look around your classroom right now, nearly every animal you can think of — your classmates, a school cat outside, a pigeon on the roof, even a small fly near the window — has **two eyes**, not one, not three. It seems so normal that most people never stop to ask: **why exactly do we need two?** Couldn't one eye in the middle of the forehead do the same job? The answer, it turns out, touches biology, physics, and several clever tests you can try at home.

② The most important reason for having two eyes is to see the world in **three dimensions**. Each of your eyes looks at the world from a slightly different position, about **six or seven centimetres** apart. When both eyes look at the same pencil, they each send a slightly different picture to the brain. The brain compares these two pictures and works out, in less than **a quarter of a second**, how far away the pencil is. This kind of depth information is called **stereopsis**, and it is almost impossible to reproduce with a single eye alone.

③ You can test stereopsis with a very simple experiment. Hold a pencil in one hand, at arm's length, and another pencil in the other hand, closer to your face. Now try to bring the tips of the two pencils together — first with **both eyes open**, and then with **one eye closed**. With both eyes, most people do it easily on the first try. With one eye closed, many people miss, sometimes by several centimetres. That is your brain reminding you that **depth is easier with two cameras than with one**.

④ A second reason for two eyes is a **wider field of view**. Each of your eyes can see a small part of the world to its own side that the other cannot. Together, they cover about **200 degrees** of the scene around your head, far more than a single eye could manage. For humans, this extra width is useful in many everyday tasks: crossing a road, catching a ball, noticing a car in the side mirror.

⑤ For many animals, the wider field of view is even more critical. Animals that hunt, such as cats, eagles and humans, usually have both eyes pointing **forward**, to give strong depth vision — useful for judging the distance to a mouse, a fish, or a ripe fruit on a tree. Animals that are often hunted, such as rabbits, horses and many birds, have eyes pointing more to the **sides**. They lose some depth vision but gain a much wider view — almost **360 degrees** in some rabbits — so that a predator cannot easily sneak up from behind.

⑥ A third important reason is **backup safety**. Eyes are delicate, and accidents do happen. A small piece of sand, a sudden bright flash, a stick while playing — any of these can briefly make one eye uncomfortable. With two eyes, we can continue to see, read, or cross a street safely even if one eye is temporarily closed. Over thousands of generations, animals with two healthy eyes have survived better than animals with only one; their children have therefore been more common.

⑦ There is also a more hidden advantage: a small amount of **eye teamwork** helps the brain improve each picture. The two eye muscles move the eyes in almost perfect coordination, many times every second, and the brain uses the differences between the two pictures to sharpen **colour, contrast and focus**. If one of your eyes works much less than the other — for example, when a child has a lazy eye that is not treated — the brain may eventually "turn off" that eye to avoid the messy signal, leading to weaker 3D vision later in life.

⑧ Of course, not every animal has exactly two "standard" eyes. **Spiders** can have up to eight simple eyes, each good at a certain job. **Scallops** have rows of tiny mirror-based eyes along the edge of their shells. **Starfish** have one small eye at the tip of each arm. These designs work for them, but for animals that run, climb, or hunt in complex 3D environments, **two forward-facing eyes** seem to be a very efficient solution that evolution has chosen many times.

⑨ So next time you close one eye on purpose — to aim at a basketball hoop, to focus a camera, or just for fun — notice how quickly the flat world feels a little **flatter**. Then open both eyes again. Depth rushes back. Two eyes, working together in silence, are quietly doing a job that one alone could never fully copy. They are, in every real sense, your **free 3D glasses** — built in, turned on, and ready from the first morning of your life.

注释： stereopsis 立体视觉；field of view 视野；predator 捕食者；coordination 协调；evolution 进化

根据文章内容，选择最佳答案。

1. What is the most important reason given for why humans have two eyes rather than one?

A. Two eyes look more beautiful than one.
B. Two eyes allow the brain to compare slightly different pictures and see the world in three dimensions (stereopsis).
C. Two eyes can blink at the same time.
D. Two eyes let us see in the dark.

2. Which simple experiment can demonstrate the value of two eyes?

A. Reading a long book out loud.
B. Closing one eye and trying to bring the tips of two pencils together.
C. Running a marathon.
D. Typing on a keyboard.

3. What is one important difference between hunters (like cats, eagles, humans) and hunted animals (like rabbits, horses)?

A. Hunters always have bigger eyes.
B. Hunters usually have both eyes pointing forward for strong depth vision, while hunted animals often have eyes on the sides for a wider view.
C. Hunted animals always have four eyes.
D. Hunters never need to see in 3D.

4. Why is having two eyes a form of backup safety, according to paragraph 6?

A. Because one eye can be sold to buy food.
B. Because, if one eye is temporarily hurt or uncomfortable, we can still continue to see, read, or cross a street safely with the other.
C. Because two eyes make us stronger in a fight.
D. Because two eyes stop all accidents from happening.

5. What is the main idea of paragraph 8?

A. All animals on Earth must have exactly two eyes.
B. Different animals, such as spiders and scallops, have very different eye designs, but for animals that run, climb or hunt in 3D, two forward-facing eyes are an efficient solution.
C. Spiders are better than humans because they have eight eyes.
D. Animals without eyes cannot survive at all.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "This kind of depth information is called stereopsis"
【解析】双眼使大脑获取深度信息实现 3D 视觉。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "Hold a pencil ... try to bring the tips of the two pencils together"
【解析】"铅笔碰铅笔"实验可体验立体视觉。A、C、D 均无关。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段 "Animals that hunt ... both eyes pointing forward ... Animals that are often hunted ... eyes pointing more to the sides"
【解析】猎食者双眼朝前深度感强，被捕食者眼侧向视野广。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段 "we can continue to see, read, or cross a street safely even if one eye is temporarily closed"
【解析】即便一眼受伤仍可靠另一眼完成日常。A、C、D 均错。故选 B。

5.
【答案】B
【题型】主旨/段落大意题
【定位】第⑧段列举蜘蛛、扇贝、海星，又指出 "two forward-facing eyes seem to be a very efficient solution"
【解析】各类动物眼睛设计多样，但复杂 3D 活动的动物更倾向前视双眼。A、C、D 均极端或片面。故选 B。